

TopoMoE: Decentralized Mixture-of-Experts (Revised Version)

Lanhaijian
Hongqiao University

Abstract

This paper revises the TopoMoE framework by integrating the macro effective duality coupling $\mathcal{C}_{\text{eff}}$ and adaptive q-deformed fusion mechanism inherited from revised Topology-GRPO. TopoMoE adopts tensor topology optimization and discrete exterior calculus for hierarchical information flow constraints. The non-commutative fusion design is philosophically inspired by Self-Entangled Duality Dynamics (SED). We explicitly state that $\mathcal{C}_{\text{eff}}$ and q-deformed operators are macroscopic engineering components, with no rigorous mathematical derivation from SED’s quantum spacetime algebra. All modules are backward compatible with the original TopoMoE and validated on decentralized multi-expert collaborative tasks.

1 TopoMoE Revised Framework

1.1 Recap of Original Core Modules

TopoMoE is a decentralized Mixture-of-Experts (MoE) architecture built on Kernelized Tensor Topology-GRPO (KT-GRPO) and discrete exterior calculus. It models experts as 0-simplices, expert interaction edges as 1-simplices, and triangular collaborative groups as 2-simplices. Four hierarchical topological constraints guarantee information flow conservation: gradient loss $\mathcal{L}_{\text{grad}}$, divergence loss $\mathcal{L}_{\text{div}}$, curl loss $\mathcal{L}_{\text{curl}}$, and curvature loss $\mathcal{L}_{\text{curv}}$. The total loss of original TopoMoE is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{grad}} + \lambda_1 \mathcal{L}_{\text{div}} + \lambda_2 \mathcal{L}_{\text{curl}} + \lambda_3 \mathcal{L}_{\text{curv}}. \quad (1)$$

1.2 KT-GRPO and Tensor Representation

For each expert e , we construct trajectory-outcome coupled tensors via Kronecker product and apply Tucker low-rank compression. The tensor-based group relative advantage is defined as:

$$A_e^{\text{KT}} = \frac{\|\mathcal{G}_e\|_F - \mu_{\mathcal{G}}}{\sigma_{\mathcal{G}}}, \quad (2)$$

where $\mathcal{G}_e$ denotes the compressed tensor core of expert e , $\mu_{\mathcal{G}}$ and $\sigma_{\mathcal{G}}$ are the mean and standard deviation of tensor core norms across all experts.

1.3 Macro Effective Duality Coupling for Experts

Motivation In decentralized MoE systems, information propagation between experts has two typical regimes: weak decoherent interaction and strong coherent closed-loop interaction. We introduce the macro effective duality coupling $\mathcal{C}_{\text{eff}}^{(e)}$ to dynamically distinguish these regimes and switch fusion strategies, fully consistent with the definition in Topology-GRPO.

Remark The macro effective duality coupling $\mathcal{C}_{\text{eff}}$ and q-deformed fusion share conceptual inspiration with SED’s duality coupling. However, they are independent macroscopic engineering parameters and operators. There is no strict mathematical derivation between SED’s quantum spacetime algebra and the present tensor-topology framework. Domains, dimensions and mathematical objects are fundamentally different.

Definition of $\mathcal{C}_{\text{eff}}^{(e)}$ For expert e , let $\tau_{\text{coh}}^{(e)}$ be the coherent reasoning steps, $\tau_{\text{delay}}^{(e)}$ be the average inter-expert message delay steps, W_{eh} be the topological weight between expert e and neighbor expert h , $\text{Var}(r_h)$ be the output variance of neighbor h , $\sigma^{\mathcal{N}(e)}$ be the standard deviation of neighbor outputs, and $\epsilon = 10^{-8}$ be the numerical stabilizer. The normalized macro effective duality coupling is:

$$\mathcal{C}_{\text{eff}}^{(e)} = \frac{\tau_{\text{coh}}^{(e)}}{\tau_{\text{delay}}^{(e)}} \cdot \frac{1}{|\mathcal{N}(e)|} \sum_{h \in \mathcal{N}(e)} W_{eh} \cdot \frac{\text{Var}(r_h)}{(\sigma^{\mathcal{N}(e)})^2 + \epsilon}. \quad (3)$$

Critical Threshold and Operating Regimes $\mathcal{C}_c$ denotes the critical coupling threshold, which can be set as a fixed hyperparameter or optimized as a learnable parameter during training. Two operating regimes are divided as follows:

- $\mathcal{C}_{\text{eff}}^{(e)} < \mathcal{C}_c$: **Weak coupling regime**. Adopt classical tensor concatenation + MLP fusion.
- $\mathcal{C}_{\text{eff}}^{(e)} \geq \mathcal{C}_c$: **Strong coherent regime**. Activate q-deformed Q-Fuse fusion.

1.4 Adaptive Tensor Message Fusion

Let $\mathbf{x}_e$ be the tensor state of expert e , and $\mathbf{msg}_e$ be the aggregated inter-expert tensor message. The adaptive fusion rule for TopoMoE is:

$$\tilde{\mathbf{x}}_e = \begin{cases} \text{CONCAT}(\mathbf{x}_e, \sigma(\text{MLP}[\mathbf{x}_e; \mathbf{msg}_e]) \odot \mathbf{msg}_e) & \mathcal{C}_{\text{eff}}^{(e)} < \mathcal{C}_c \\ \text{Q-Fuse}(\mathbf{x}_e, \mathbf{msg}_e; \tilde{\theta}) & \mathcal{C}_{\text{eff}}^{(e)} \geq \mathcal{C}_c \end{cases} \quad (4)$$

1.5 Q-Fuse: q-Deformed Non-Commutative Fusion

We adopt the unified Q-Fuse definition consistent with Topology-GRPO. Let $\phi(\cdot) = W_\phi \cdot + b_\phi$ and $\psi(\cdot) = W_\psi \cdot + b_\psi$ be learnable linear maps:

$$\text{Q-Fuse}(\mathbf{x}_e, \mathbf{msg}_e; \tilde{\theta}) = W_1 \mathbf{x}_e + W_2 \mathbf{msg}_e + \cos(2\pi\tilde{\theta}) \cdot (\phi(\mathbf{x}_e) \odot \psi(\mathbf{msg}_e) - \phi(\mathbf{msg}_e) \odot \psi(\mathbf{x}_e)). \quad (5)$$

Non-commutativity verification:

$$\text{Q-Fuse}(\mathbf{msg}_e, \mathbf{x}_e) - \text{Q-Fuse}(\mathbf{x}_e, \mathbf{msg}_e) = 2 \cos(2\pi\tilde{\theta}) (\phi(\mathbf{msg}_e) \odot \psi(\mathbf{x}_e) - \phi(\mathbf{x}_e) \odot \psi(\mathbf{msg}_e)) \neq 0. \quad (6)$$

When $\tilde{\theta} \rightarrow 0$ or the system falls into the weak coupling regime, the non-commutative term vanishes, and Q-Fuse degenerates to classical linear fusion.

1.6 Adaptive Update for $\tilde{\theta}$

The twist angle $\tilde{\theta}$ is optimized independently of tensor and expert networks. The update is only activated in the strong coherent regime, with the same rule as Topology-GRPO:

$$\tilde{\theta}(t+1) = \tilde{\theta}(t) + \eta_\theta \cdot \mathbb{I}_{\{\mathcal{C}_{\text{eff}}^{(e)} \geq \mathcal{C}_c\}} \cdot (\mathcal{C}_{\text{eff}}^{(e)} - \mathcal{C}_c) \cdot \frac{\partial \|\text{Q-Fuse}(\mathbf{x}_e, \mathbf{msg}_e) - \text{Q-Fuse}(\mathbf{msg}_e, \mathbf{x}_e)\|^2}{\partial \tilde{\theta}}. \quad (7)$$

where η_θ is the learning rate for $\tilde{\theta}$.

1.7 Dynamical Topology Constraint Weights

Original four topological constraints use fixed weights. We extend them to $\mathcal{C}_{\text{eff}}$ -adaptive weights to match different coupling regimes:

$$\begin{cases} \lambda_1 = \lambda_1^0, \lambda_2 = \lambda_2^0, \lambda_3 = \lambda_3^0, & \mathcal{C}_{\text{eff}}^{(e)} < \mathcal{C}_c \\ \lambda_1 = \lambda_1^0, \lambda_2 = \lambda_2^0 \cdot (1 + \gamma \cdot \mathcal{C}_{\text{eff}}^{(e)}), \lambda_3 = \lambda_3^0 \cdot (1 + \gamma \cdot \mathcal{C}_{\text{eff}}^{(e)}), & \mathcal{C}_{\text{eff}}^{(e)} \geq \mathcal{C}_c \end{cases} \quad (8)$$

where $\lambda_1^0, \lambda_2^0, \lambda_3^0$ are initial fixed weights, and $\gamma > 0$ is a scaling factor. In strong coherent regimes, we enhance curl and curvature constraints to suppress cyclic logical conflicts and global manifold deviation.

Note: λ_1 remains fixed as λ_1^0 because divergence conservation (local information balance) is a first-order constraint that should hold regardless of coupling regime.

1.8 Updated Total Loss Function

Combining original topological constraints with adaptive dynamic weights, the final total loss is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{grad}} + \lambda_1^0 \mathcal{L}_{\text{div}} + \lambda_2(\mathcal{C}_{\text{eff}}) \mathcal{L}_{\text{curl}} + \lambda_3(\mathcal{C}_{\text{eff}}) \mathcal{L}_{\text{curv}}. \quad (9)$$

2 Training Algorithm

Algorithm 1 Updated TopoMoE Training Algorithm

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1: Initialize expert networks, tensor encoders and decentralized message modules
2: Initialize hyperparameters:  $\lambda_1^0, \lambda_2^0, \lambda_3^0, \gamma, \eta_\theta$ 
3: Initialize critical threshold  $\mathcal{C}_c$  (fixed or learnable), initial  $\tilde{\theta}^{(0)}$ 
4: while training not converged do
5:   Construct simplicial complex: 0-1-2-simplices for experts and interactions
6:   Collect reasoning trajectories and tensor features for all experts
7:   for each expert  $e$  do
8:     Compute  $\tau_{\text{coh}}^{(e)}, \tau_{\text{delay}}^{(e)}, \text{Var}(r_h), \sigma^{\mathcal{N}(e)}$ 
9:     Calculate  $\mathcal{C}_{\text{eff}}^{(e)}$  via Eq. (3)
10:    Obtain aggregated tensor message  $\text{msg}_e$  via message passing
11:    Compute fused tensor state  $\tilde{\mathbf{x}}_e$  via Eq. (4)
12:    Calculate tensor advantage  $A_e^{\text{KT}}$  via Eq. (2)
13:    Update dynamic constraint weights  $\lambda_2, \lambda_3$  via Eq. (8) ( $\lambda_1 = \lambda_1^0$  fixed)
14:  end for
15:  Compute total loss  $\mathcal{L}_{\text{total}}$  via Eq. (9)
16:  Update all expert and encoder networks
17:  if  $\exists e, \mathcal{C}_{\text{eff}}^{(e)} \geq \mathcal{C}_c$  then
18:    Update  $\tilde{\theta}$  via Eq. (7)
19:  end if
20: end while
21: return All expert networks and tensor encoders

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3 Compatibility and Ablation Study

3.1 Backward Compatibility

The revised TopoMoE is fully backward compatible with the original version. When $\mathcal{C}_{\text{eff}}^{(e)} \rightarrow 0$ or $\tilde{\theta} \rightarrow 0$, the adaptive fusion and dynamic weights degenerate to the original fixed-parameter framework. All existing datasets and experimental pipelines can be reused directly.

3.2 Ablation Design

To evaluate the contribution of each newly added module, we design the ablation groups consistent with Topology-GRPO:

1. Baseline: Original TopoMoE (no $\mathcal{C}_{\text{eff}}$, no Q-Fuse, fixed constraint weights)
2. $\mathcal{C}_{\text{eff}}$ only: Adaptive threshold enabled, always use classical fusion
3. Full model: $\mathcal{C}_{\text{eff}}$ + Q-Fuse + decoupled $\tilde{\theta}$ + dynamic constraint weights
4. Ablate non-commutativity: Set $\cos(2\pi\tilde{\theta}) \equiv 0$
5. Fixed $\mathcal{C}_c$ vs. learnable $\mathcal{C}_c$

4 Cross-Framework Consistency

This revision maintains full consistency across the whole system:

- Unified symbol system: $\mathcal{C}_{\text{eff}}, \mathcal{C}_c, \tilde{\theta}$, Q-Fuse share identical definitions;
- Unified physical logic: weak coupling (classical fusion) / strong coupling (non-commutative fusion);
- Unified training mechanism: decoupled $\tilde{\theta}$ update activated only in strong coherent regimes;
- Unified academic statement: conceptual inspiration from SED, no strict mathematical derivation.